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Fabric with electrical components

Abstract

One or more electrical components may be incorporated into a piece of fabric. The electrical component may include an internal portion that is located inside of the fabric, an external portion that is located on an exterior surface of the fabric, and protrusions that extend through the fabric to electrically and/or mechanically couple the internal and external portions of the electrical component. The internal portion of the component may be inserted into the fabric during formation of the fabric. The external portion of the component may be coupled to the internal portion after the fabric is formed by inserting the protrusions on the internal portion into recesses in the external portion. The external portion of the component may contain skin-facing and/or viewer-facing input-output devices, while the internal portion may contain circuitry that electrically communicates with the input-output devices in the external portion.

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Background/Summary

(1) This application claims the benefit of provisional patent application No. 63/238,615, filed Aug. 30, 2021, which is hereby incorporated by reference herein in its entirety.

FIELD

(1) This relates generally to items with fabric and, more particularly, to items with fabric and electrical components.

BACKGROUND

(2) It may be desirable to form bags, furniture, clothing, and other items from materials such as fabric. Fabric items generally do not include electrical components. It may be desirable, however, to incorporate electrical components into fabric to provide a user of a fabric item with enhanced functionality.

(3) It can be challenging to incorporate electrical components into fabric. Fabric is flexible, so it can be difficult to mount structures to fabric. Electrical components must be coupled to signal paths (e.g., signal paths that carry data signals, power, etc.), but unless care is taken, signal paths may be damaged, or components may become dislodged as fabric is bent or stretched.

(4) It would therefore be desirable to be able to provide improved techniques for incorporating electrical components into items with fabric.

SUMMARY

(5) Interlacing equipment (e.g., weaving equipment, knitting equipment, braiding equipment, etc.) may be provided with individually adjustable components. The use of individually adjustable components may allow electrical components to be inserted into and/or embedded in the fabric during the creation or formation of the fabric.

(6) The interlacing equipment may create a gap between first and second fabric portions during interlacing operations. The gap may be a void between fabric portions or the gap may be a position or location between fabric portions. An insertion tool may insert an electrical component into the gap, and the electrical component may be electrically coupled to conductive strands in the gap, if desired.

(7) An electrical component may include an internal portion that is located inside of the fabric, an external portion that is located on an exterior surface of the fabric, and protrusions that extend through the fabric to electrically and/or mechanically couple the internal and external portions of the electrical component. The internal portion of the component may be inserted into the fabric during formation of the fabric. After the fabric is formed, protrusions on the internal portion may extend through the fabric and may be exposed to the exterior of the fabric. The external portion of the component may then be coupled to the internal portion by attaching the protrusions to mating attachment structures (e.g., recesses) on the external portion of the fabric.

(8) The external portion of the component may contain skin-facing and/or viewer-facing input-output devices (e.g., a heart-rate sensor, a blood oxygen sensor, a microphone, a speaker, a display, an indicator, a touch screen, a button, etc.), while the internal portion may contain circuitry that electrically communicates with the input-output device(s) in the external portion.

(9) The fabric may be used to form a wrist strap or other wearable item.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of an illustrative fabric item in accordance with an embodiment.

(2) FIG. 2 is a side view of illustrative fabric in accordance with an embodiment.

(3) FIG. 3 is a side view of layers of material that may be incorporated into a fabric item in

accordance with an embodiment.

(4) FIG. 4 is a diagram illustrating how interlacing equipment may be used to create fabric while an insertion tool is used to insert one or more electrical components into the fabric in accordance with an embodiment.

(5) FIG. 5 is a cross-sectional side view of an illustrative electrical component in accordance with an embodiment.

(6) FIG. 6 is a cross-sectional side view of an illustrative electrical component having an electrical device mounted on an interconnect substrate in accordance with an embodiment.

(7) FIG. 7 is a cross-sectional side view of an illustrative electrical component having a protective structure in accordance with an embodiment.

(8) FIG. 8 is a cross-sectional side view of an illustrative electrical component having recesses for receiving strands in accordance with an embodiment.

(9) FIG. 9 is a cross-sectional side view of an illustrative electrical component having an internal portion located within a piece of fabric and having protrusions that protrude out of the fabric in accordance with an embodiment.

(10) FIG. 10 is a cross-sectional side view of an illustrative electrical component of the type shown in FIG. 9 with an external portion located outside of the fabric in accordance with an embodiment.

(11) FIG. 11 is a cross-sectional side view of an illustrative electrical component having a protective encapsulant around an external portion in accordance with an embodiment.

(12) FIG. 12 is a top view of a protective encapsulant of the type shown in FIG. 11 surrounding an external portion of an electrical component in accordance with an embodiment.

(13) FIG. 13 is a cross-sectional side view of an illustrative electrical component having protrusions with spherical ends in accordance with an embodiment.

(14) FIG. 14 is a cross-sectional side view of an illustrative electrical component having protrusions with T-shaped ends in accordance with an embodiment.

(15) FIG. 15 is a cross-sectional side view of an illustrative electrical component having protrusions with L-shaped ends in accordance with an embodiment.

(16) FIG. 16 is a top view of an illustrative external portion of an electrical component having recesses for receiving protrusions on an internal portion of the electrical component in accordance with an embodiment.

(17) FIG. 17 is a cross-sectional side view of an illustrative electrical component having an internal portion located within a piece of fabric and coupled to first and second external portions located outside of the fabric in accordance with an embodiment.

(18) FIG. 18 is a top view of an illustrative electronic device such as a watch having a fabric strap that incorporates one or more electrical components in accordance with an embodiment.

(19) FIG. 19 is a cross-sectional side view of an illustrative electronic device of the type shown in FIG. 18 with a fabric strap that incorporates an electrical component having an internal portion located within the fabric and an external portion located outside of the fabric in accordance with an embodiment.

DETAILED DESCRIPTION

(20) Electronic devices, enclosures, and other items may be formed from fabric such as woven fabric, knit fabric, or other suitable fabric. The fabric may include strands of insulating and conductive material. Conductive strands may form signal paths through the fabric and may be coupled to electrical components such as light-emitting diodes and other light-emitting devices, integrated circuits, sensors, haptic output devices, and other circuitry.

(21) Interlacing equipment (sometimes referred to as intertwining equipment) may include weaving equipment, knitting equipment, braiding equipment, or any other suitable equipment used for crossing, looping, overlapping, or otherwise coupling strands of material together to form a network of strands (e.g., fabric). Interlacing equipment may be provided with individually adjustable components such as warp strand positioning equipment (e.g., heddles or other warp

strand positioning equipment), weft strand positioning equipment, a reed, take-down equipment, let off equipment (e.g., devices for individually dispensing and tensioning warp strands), needle beds, feeders, guide bars, strand processing and component insertion equipment, and other components for forming fabric items. The individual adjustability of these components may allow interlacing operations (e.g., weaving operations, knitting operations, braiding operations, and/or other interlacing operations) to be performed without requiring continuous lock-step synchronization of each of these devices, thereby allowing fabric with desired properties to be woven. As an example, normal reed movement and other weaving operations may be periodically suspended and/or may periodically be out-of-sync with other components to accommodate component insertion operations whereby electrical components (sometimes referred to as nodes or smart nodes) are inserted into the fabric during the creation or formation of the fabric.

(22) Items such as item **10** of FIG. **1** may include fabric and may sometimes be referred to as a fabric item or fabric-based item. Item **10** may be an electronic device or an accessory for an electronic device such as a laptop computer, a computer monitor containing an embedded computer, a tablet computer, a cellular telephone, a media player, or other handheld or portable electronic device, a smaller device such as a wrist-watch device, a pendant device, a headphone or earpiece device, a device embedded in eyeglasses or other equipment worn on a user's head, or other wearable or miniature device, a television, a computer display that does not contain an embedded computer, a gaming device, a navigation device, an embedded system such as a system in which fabric item **10** is mounted in a kiosk, in an automobile, airplane, or other vehicle (e.g., an autonomous or non-autonomous vehicle), other electronic equipment, or equipment that implements the functionality of two or more of these devices. If desired, item **10** may be a removable external case for electronic equipment, may be a strap, may be a wrist band or head band, may be a removable cover for a device, may be a case or bag that has straps or that has other structures to receive and carry electronic equipment and other items, may be a necklace or arm band, may be a wallet, sleeve, pocket, or other structure into which electronic equipment or other items may be inserted, may be part of a chair, sofa, or other seating (e.g., cushions or other seating structures), may be part of an item of clothing or other wearable item (e.g., a hat, belt, wrist band, headband, etc.), or may be any other suitable item that incorporates fabric.

(23) Item **10** may include interlaced strands of material such as monofilaments and yarns that form fabric **12**. As used herein, “interlaced” strands of material and “intertwined” strands of material may both refer to strands of material that are crossed with one another, looped with one another, overlapping one another, or otherwise coupled together (e.g., as part of a network of strands that make up a fabric). Fabric **12** may form all or part of a housing wall or other layer in an electronic device, may form internal structures in an electronic device, or may form other fabric-based structures. Item **10** may be soft (e.g., item **10** may have a fabric surface that yields to a light touch), may have a rigid feel (e.g., the surface of item **10** may be formed from a stiff fabric), may be coarse, may be smooth, may have ribs or other patterned textures, and/or may be formed as part of a device that has portions formed from non-fabric structures of plastic, metal, glass, crystalline materials, ceramics, or other materials.

(24) The strands of material used in forming fabric **12** may be single-filament strands (sometimes referred to as fibers) or may be threads, yarns, or other strands that have been formed by interlacing multiple filaments of material together. Strands may be formed from polymer, metal, glass, graphite, ceramic, natural materials such as cotton or bamboo, or other organic and/or inorganic materials and combinations of these materials. Conductive coatings such as metal coatings may be formed on non-conductive strands (e.g., plastic cores) to make them conductive. Reflective coatings such as metal coatings may be applied to strands to make them reflective. Strands may also be formed from single-filament metal wire (e.g., bare metal wire), multifilament wire, or combinations of different materials. Strands may be insulating or conductive.

(25) Strands in fabric **12** may be conductive along their entire lengths or may have conductive

portions. Strands may have metal portions that are selectively exposed by locally removing insulation (e.g., to form connections with other conductive strand portions and/or to form connections with electrical components). Strands may also be formed by selectively adding a conductive layer to a portion of a non-conductive strand). Threads and other multifilament yarns that have been formed from interlaced filaments may contain mixtures of conductive strands and insulating strands (e.g., metal strands or metal coated strands with or without exterior insulating layers may be used in combination with solid plastic strands or natural strands that are insulating). In some arrangements, which may sometimes be described herein as an example, fabric **12** may be a woven fabric and the strands that make up fabric **12** may include warp strands and weft strands.

(26) Conductive strands and insulating strands may be woven, knit, or otherwise interlaced to form conductive paths. The conductive paths may be used in forming signal paths (e.g., signal buses, power lines for carrying power, etc.), may be used in forming part of a capacitive touch sensor electrode, a resistive touch sensor electrode, or other input-output device, or may be used in forming other patterned conductive structures. Conductive structures in fabric **12** may be used in carrying electrical current such as power, digital signals, analog signals, sensor signals, control signals, data, input signals, output signals, or other suitable electrical signals.

(27) Item **10** may include additional mechanical structures **14** such as polymer binder to hold strands in fabric **12** together, support structures such as frame members, housing structures (e.g., an electronic device housing), and other mechanical structures.

(28) To enhance mechanical robustness and electrical conductivity at strand-to-strand connections and/or strand-to-component connections, additional structures and materials (e.g., solder, crimped metal connections, welds, conductive adhesive such as anisotropic conductive film and other conductive adhesive, non-conductive adhesive, fasteners, etc.) may be used in fabric **12**. Strand-to-strand connections may be formed where strands cross each other perpendicularly or at other strand intersections where connections are desired. Insulating material can be interposed between intersecting conductive yarns at locations in which it is not desired to form a strand-to-strand connection. The insulating material may be plastic or other dielectric, may include an insulating strand or a conductive strand with an insulating coating or insulated conductive monofilaments, etc. Solder connections may be formed between conductive strands and/or between conductive strands and electrical components by melting solder so that the solder flows over conductive strands. The solder may be melted using an inductive soldering head to heat the solder, using hot air to heat the solder, using a reflow oven to heat the solder, using a laser or hot bar to heat the solder, or using other soldering equipment. In some arrangements, outer dielectric coating layers (e.g., outer polymer layers) may be melted away in the presence of molten solder, thereby allowing underlying metal yarns to be soldered together. In other arrangements, outer dielectric coating layers may be removed prior to soldering (e.g., using laser ablation equipment or other coating removal equipment).

(29) Circuitry **16** may be included in item **10**. Circuitry **16** may include electrical components that are coupled to fabric **12**, electrical components that are housed within an enclosure formed by fabric **12**, electrical components that are attached to fabric **12** using welds, solder joints, adhesive bonds (e.g., conductive adhesive bonds such as anisotropic conductive adhesive bonds or other conductive adhesive bonds), crimped connections, or other electrical and/or mechanical bonds. Circuitry **16** may include metal structures for carrying current, electrical components such as integrated circuits, light-emitting diodes, sensors, and other electrical devices. Control circuitry in circuitry **16** may be used to control the operation of item **10** and/or to support communications with item **18** and/or other devices.

(30) Item **10** may interact with electronic equipment or other additional items **18**. Items **18** may be attached to item **10** or item **10** and item **18** may be separate items that are configured to operate with each other (e.g., when one item is a case and the other is a device that fits within the case, etc.). Circuitry **16** may include antennas and other structures for supporting wireless

communications with item **18**. Item **18** may also interact with item **10** using a wired communications link or other connection that allows information to be exchanged.

(31) In some situations, item **18** may be an electronic device such as a cellular telephone, computer, or other portable electronic device and item **10** may form a cover, case, bag, or other structure that receives the electronic device in a pocket, an interior cavity, or other portion of item **10**. In other situations, item **18** may be a wrist-watch device or other electronic device and item **10** may be a strap or other fabric item that is attached to item **18** (e.g., item **10** and item **18** may together form a fabric-based item such as a wristwatch with a strap). In still other situations, item **10** may be an electronic device, fabric **12** may be used in forming the electronic device, and additional items **18** may include accessories or other devices that interact with item **10**. Signal paths formed from conductive yarns and monofilaments may be used to route signals in item **10** and/or item(s) **18**.

(32) The fabric that makes up item **10** may be formed from yarns and/or monofilaments that are interlaced using any suitable interlacing equipment. With one suitable arrangement, which may sometimes be described herein as an example, fabric **12** may be woven fabric formed using a weaving machine. In this type of illustrative configuration, fabric may have a plain weave, a basket weave, a satin weave, a twill weave, or variations of these weaves, may be a three-dimensional woven fabric, or may be other suitable fabric. This is, however, merely illustrative. If desired, fabric **12** may include knit fabric, warp knit fabric, weft knit fabric, braided fabric, other suitable type of fabric, and/or a combination of any two or more of these types of fabric.

(33) A cross-sectional side view of illustrative woven fabric **12** is shown in FIG. **2**. As shown in FIG. **2**, fabric **12** may include strands **80**. Strands **80** may include warp strands **20** and weft strands **22**. If desired, additional strands that are neither warp nor weft strands may be incorporated into fabric **12**. The example of FIG. **2** is merely illustrative. In the illustrative configuration of FIG. **2**, fabric **12** has a single layer of woven strands **80**. Multi-layer fabric constructions may be used for fabric **12** if desired.

(34) Item **10** may include non-fabric materials (e.g., structures formed from plastic, metal, glass, ceramic, crystalline materials such as sapphire, etc.). These materials may be formed using molding operations, extrusion, machining, laser processing, and other fabrication techniques. In some configurations, some or all of item **10** may include one or more layers of material such as layers **24** of FIG. **3**. Layers **24** may include layers of polymer, metal, glass, fabric, adhesive, crystalline materials, ceramic, substrates on which components have been mounted, patterned layers of material, layers of material containing patterned metal traces, thin-film devices such as transistors, and/or other layers.

(35) A diagram illustrating how electrical components may be inserted into fabric **12** during the formation of fabric **12** is illustrated in FIG. **4**. As shown in FIG. **4**, fabric **12** may be formed from fabric portions such as fabric portions **12-1** and **12-2**. Fabric portions **12-1** and **12-2** may be formed from interlaced strands **80**. For example, a first set of strands **80** may be used to form fabric portion **12-1** and a second set of strands **80** may be used to form fabric portion **12-2**. Fabric portions **12-1** and **12-2** may be different portions of a single layer of fabric **12**, or fabric portion **12-1** may form one or more first layers of fabric **12** and fabric portion **12-2** may form one or more second layers of fabric **12**.

(36) Using interlacing equipment **120**, strands **80** may be interlaced to form fabric **12**. Interlacing equipment **120** may be weaving equipment, knitting equipment, braiding equipment, or other suitable interlacing equipment. Interlacing equipment **120** may be used to create one or more regions in fabric **12** such as pocket **66** (sometimes referred to as a gap, space, cavity, void, position, location, etc.) for receiving electrical components. Regions in fabric **12** that receive electrical components such as pocket **66** may be formed by creating a space or gap between portions of fabric **12** such as fabric portion **12-1** and fabric portion **12-2**. The term “pocket” may be used to refer to a void between fabric portions and/or may be used to refer to a position or location between fabric portions (e.g., a position between strands of material in fabric **12**, with or without an actual void).

(37) Electrical components may be inserted into pocket **66** during the formation of fabric **12** using component insertion equipment such as insertion tool **54**. Insertion tool **54** may hold component **26** and may position component **26** in pocket **66** during interlacing operations (e.g., by moving component **26** towards pocket **66** in direction **140**). If desired, component **26** may be electrically and/or mechanically connected to one or more conductive strands **80C** in pocket **66**. Following insertion and attachment of component **26**, interlacing equipment **120** may continue interlacing operations (which may include closing pocket **66**, if desired) to continue forming fabric **12**.

(38) In some arrangements, processing steps such as alignment of component **26** with conductive strands **80C**, electrically connecting (e.g., soldering) component **26** to conductive strands **80C**, encapsulation of the electrical connection between component **26** and conductive strands **80C**, and/or verification of the integrity of the electrical connection between component **26** and conductive strands **80C** may be performed after component **26** is inserted into pocket **66**.

(39) In some arrangements, the gap between first and second fabric portions **12-1** and **12-2** may remain in place after electrical component **26** is enclosed in fabric **12** (e.g., a space may exist between fabric portions **12-1** and **12-2** after formation of fabric **12** is complete). In other arrangements, first and second fabric portions **12-1** and **12-2** may be pulled together such that gap **66** is eliminated after electrical component **26** is enclosed in the gap (e.g., fabric portions **12-1** and **12-2** may be in contact with one another without an intervening gap after the formation of fabric **12** is complete). Fabric **12** may have a bulge where electrical component **26** is located, or fabric **12** may not have a bulge where electrical component **26** is located (e.g., the fabric may have substantially uniform thickness across locations with electrical components **26** and locations without electrical components **26**, if desired).

(40) A side view of an illustrative electrical component of the type that may be used in item **10** is shown in FIG. **5**. Electrical components in item **10** such as illustrative electrical component **26** of FIG. **5** may include discrete electrical components such as resistors, capacitors, and inductors, may include connectors, may include batteries, may include input-output devices such as switches, buttons, light-emitting components such as light-emitting diodes, audio components such as microphones and speakers, vibrators (e.g., piezoelectric actuators that can vibrate), solenoids, electromechanical actuators, motors, and other electromechanical devices, microelectromechanical systems (MEMS) devices, pressure sensors, light detectors, proximity sensors (light-based proximity sensors, capacitive proximity sensors, etc.), force sensors (e.g., piezoelectric force sensors), strain gauges, moisture sensors, temperature sensors, accelerometers, gyroscopes, compasses, magnetic sensors (e.g., Hall effect sensors and magnetoresistance sensors such as giant magnetoresistance sensors), touch sensors, and other sensors, components that form displays, touch sensor arrays (e.g., arrays of capacitive touch sensor electrodes to form a touch sensor that detects touch events in two dimensions), and other input-output devices, energy storage devices, electrical components that form control circuitry such as non-volatile and volatile memory, microprocessors, application-specific integrated circuits, system-on-chip devices, baseband processors, wired and wireless communications circuitry, and other integrated circuits.

(41) Electrical components such as component **26** may be bare semiconductor dies (e.g., laser dies, light-emitting diode dies, integrated circuits, etc.) or packaged components (e.g. semiconductor dies or other devices packaged within plastic packages, ceramic packages, or other packaging structures). One or more electrical terminals such as contact pads **30** may be formed on body **28** of component **26**. Body **28** (sometimes referred to as device **28**, electrical device **28**, etc.) may be a semiconductor die (e.g., a laser die, light-emitting diode die, integrated circuit, etc.) or may be a package for a component (e.g., a plastic package or other dielectric package that contains one or more semiconductor dies or other electrical devices). Contacts for body **28** such as pads **30** may be protruding leads, may be planar contacts, may be formed in an array, may be formed on any suitable surfaces of body **28**, or may be any other suitable contacts for forming electrical connections to component **26**. For example, pads **30** may be metal solder pads.

(42) As shown in the example of FIG. 6, body **28** may be mounted on a support structure such as substrate **36**. Interposer **36** (sometimes referred to as an interconnect substrate, a printed circuit substrate, etc.) may be a printed circuit, ceramic carrier, or other substrate. The layer(s) forming interconnect substrate **36** may include one or more flexible printed circuit layers such as polyimide layers, one or more layers of rigid printed circuit board material such as fiberglass-filled epoxy (e.g., FR4), and/or layers of other materials (e.g., other dielectric materials such as silicone, other elastomeric material, other flexible polymers, etc.). Interconnect substrate **36** may be larger than body **28** or may have other suitable sizes. Interconnect substrate **36** may have a planar shape with a thickness of 700 microns, more than 500 microns, less than 500 microns, or other suitable thickness. The thickness of body **28** may be 500 microns, more than 300 microns, less than 1000 microns, or other suitable thickness. The footprint (area viewed from above) of body **28** and substrate **36** may be 10 microns×10 microns, 100 microns×100 microns, more than 1 mm×1 mm, less than 10 mm×10 mm, may be rectangular, may be square, may have L-shapes, or may have other suitable shapes and sizes.

(43) Interconnect substrate **36** may contain signal paths such as metal traces **38**. Metal traces **38** (sometimes referred to as interconnects, signal paths, etc.) may have portions forming contacts such as pads **34** and **40**. Pads **34** and **40** may be formed on the upper surface of interconnect substrate **36**, on the lower surface of interconnect substrate **36**, and/or on the sides of interconnect substrate **36**. Conductive material such as conductive material **32** may be used in mounting body **28** to interconnect substrate **36**. Conductive material **32** may be solder (e.g., low temperature solder, high temperature solder, etc.), may be conductive adhesive (isotropic conductive adhesive or anisotropic conductive film), may be formed during welding, and/or may be other conductive material for coupling electrical device pads (body pads) such as pads **30** on body **28** to interconnect substrate pads **34**. Metal traces **38** in substrate **36** may couple pads **34** to other pads such as pads **40**. If desired, pads **40** may be larger and/or more widely spaced than pads **34**, thereby facilitating attachment of substrate **36** to conductive yarns and/or other conductive paths in item **10**. Solder, conductive adhesive, or other conductive connections may be used in coupling pads **40** to conductive strands, printed circuit traces, or other conductive path materials in item **10**.

(44) FIG. 7 shows an example in which component **26** includes a protective structure such as protective structure **130** on interconnect substrate **36**. Protective structure **130** may, for example, be a plastic structure that completely or partially encapsulates devices **28** and interconnect substrate **36** to provide mechanical robustness, protection from moisture and other environmental contaminants, heat sinking, and/or electrical insulation. Protective structure **130** may be formed from molded plastic (e.g., injection-molded plastic, insert molded plastic, transfer molded plastic, low-pressure molded plastic, two-part molded plastic, etc.) that has been molded over one or more devices **28** and substrate **36** or that is molded into the desired shape and subsequently attached to substrate **36**, may be a layer of encapsulant material (e.g., thermoplastic) that has been melted to encapsulate devices **28**, may be a layer of polymer such as polyimide that has been cut or machined into the desired shape and subsequently attached to substrate **36**, or may be formed using other suitable methods. Illustrative materials that may be used to form protective structure **130** include epoxy, polyamide, polyurethane, silicone, thermoplastic, other suitable materials, or a combination of any two or more of these materials. Protective structure **130** may be formed on one or both sides of substrate **36** (e.g., may completely or partially surround substrate **36**).

(45) Protective structure **130** may be entirely opaque, may be entirely transparent, or may have both opaque and transparent regions. Transparent portions of protective structure **130** may allow light emitted from one or more devices **28** to be transmitted through protective structure **130** and/or may allow external light to reach (and be detected by) one or more devices **28**. If desired, one or more openings, recesses, grooves, and/or other features may be formed in protective structure **130**. For example, an opening may be formed in protective structure **130** to allow light to be detected by and/or emitted from one or more devices **28**. Protective structure **130** may include one or more

grooves for receiving strands (e.g., conductive or insulating strands) in fabric **12**.

(46) Protective structure **130** may, if desired, have different thicknesses. The example of FIG. 7 in which protective structure **130** has uniform thickness across substrate **36** is merely illustrative. In some arrangements, protective structure **130** may be an encapsulant material such as thermoplastic that has been melted to create a robust connection between component **26** and strands **80** of fabric **12**. For example, protective structure **130** may surround portions of strands **80**, may fill recesses, grooves, or other features in component **26** to help interlock component **26** to strands **80**, and/or may fill gaps in fabric **12**. Protective structure **130** may include one or more different types of materials, if desired (e.g., one or more different thermoplastic materials with different melting temperatures).

(47) If desired, substrate **36** may be sufficiently large to accommodate multiple electrical devices each with a respective body **28**. For example, one or more light-emitting diodes, sensors, microprocessors, and/or other electrical devices may be mounted to a common substrate such as substrate **36** of FIG. 7. The light-emitting diodes may be micro-light-emitting diodes (e.g., light-emitting diode semiconductor dies having footprints of about 10 microns×10 microns, more than 5 microns×5 microns, less than 100 microns×100 microns, or other suitable sizes). The light-emitting diodes may include light-emitting diodes of different colors (e.g., red, green, blue, white, etc.), infrared light, or ultraviolet light. Redundant light-emitting diodes or other redundant circuitry may be included on substrate **36**. In configurations of the type shown in FIG. 7 in which multiple electrical devices (each with a respective body **28**) are mounted on a common substrate, electrical component **26** may include any suitable combination of electrical devices (e.g., light-emitting diodes, sensors, integrated circuits, actuators, energy storage devices, and/or other devices of the type described in connection with electrical component **26** of FIG. 5).

(48) The examples of FIGS. 6 and 7 in which devices **28** are only located on one side of substrate **36** are merely illustrative. If desired, devices **28** may be mounted to both sides of substrate **36**.

(49) Electrical components **26** may be coupled to fabric structures, individual strands, printed circuits (e.g., rigid printed circuits formed from fiberglass-filled epoxy or other rigid printed circuit board material or flexible printed circuits formed from polyimide substrate layers or other sheets of flexible polymer materials), metal or plastic parts with signal traces, or other structures in item **10**.

(50) In some configurations, item **10** may include electrical connections between components **26** and conductive paths in fabric **12**. As shown in FIG. 8, for example, component **26** may be coupled to conductive strands **80C** of fabric **12**. Conductive strands **80C** (sometimes referred to as “wires”) may be configured to carry electrical signals (e.g., power, digital signals, analog signals, sensor signals, control signals, data signals, input signals, output signals, or other suitable electrical current) to and/or from components **26**. Strands **80C** may be warp strands (e.g., warp strands **20** of FIG. 2), weft strands (e.g., weft strands **22** of FIG. 2), or other suitable strands **80** in fabric **12**. If desired, component **26** may be coupled to only a single conductive strand **80C**, may be coupled to two conductive strands **80C**, or may be coupled to three or more conductive strands **80C**. If desired, component **26** may also or instead be coupled to insulating strands in fabric **12**. Arrangements in which component **26** is coupled to a pair of conductive strands **80C** are sometimes described herein as an illustrative example.

(51) Component **26** may have contact pads such as pads **40**. Conductive material **82** may be used to couple pads **40** to conductive strands **80C**. Conductive material **82** may be solder, anisotropic conductive adhesive, or other conductive material. Arrangements in which conductive material **82** is formed from solder may sometimes be described herein as an illustrative example. In the example of FIG. 8, pads **40** are formed on the same surface of substrate **36** on which device **28** is mounted. Conductive material **82** may be used to electrically and mechanically couple component **26** to strands **80C** of fabric **12**. If desired, pads **40** may also or instead be additionally formed on the lower surface of substrate **36** (e.g., the surface opposite the surface on which device **28** is mounted). The example of FIG. 8 is merely illustrative.

(52) In some configurations, it may be desirable to provide a more robust mechanical connection between component **26** and fabric **12** to ensure that component **26** does not come loose when fabric **12** is bent or stretched. To increase the robustness of the connection between strands **80C** and component **26**, component **26** may have one or more recesses for receiving strands **80C**. For example, one or more strands **80** may be threaded through a portion of component **26** to help secure component **26** to fabric **12**. Strands **80** may be threaded through openings (sometimes referred to as recesses, trenches, grooves, holes, slots, notches, etc.) of component **26**. The openings may be formed in device **28**, interconnect substrate **36**, protective structure **130**, and/or other portions of component **26**. FIG. **8** shows an example in which conductive strands **80C** are received within grooves such as grooves **50** that are formed in protective structure **130**. This is, however, merely illustrative. If desired, grooves **50** may instead or additionally be formed in interconnect substrate **36**, device **28**, and/or other portions of component **26**. The location, shape, and geometry of grooves **50** of FIG. **8** are merely illustrative.

(53) Grooves **50** (sometimes referred to as recesses, trenches, openings, holes, slots, notches, etc.) in protective structure **130** may be formed by removing portions of protective structure **130** (e.g., using a laser, a mechanical saw, a mechanical mill, or other equipment) or may be formed by molding (e.g., injection molding, insert molding, etc.) or otherwise forming protective structure **130** into a shape that includes grooves **50**. Grooves **50** may have a width between 2 mm and 6 mm, between 0.3 mm and 1.5 mm, between 1 mm and 5 mm, between 3 mm and 8 mm, greater than 3 mm, less than 3 mm, or other suitable width. If desired, grooves **50** may have different depths (e.g., to expose contact pads **40** that are located at different surface heights of interconnect substrate **36**).

(54) In the example of FIG. **8**, grooves **50** expose conductive pads **40** on interconnect substrate **36**. Strands **80C** may each be threaded through an associated one of grooves **50** in protective structure **130**. Solder or other conductive material **82** may be used to electrically and mechanically couple strands **80C** to conductive pads **40** in grooves **50** of protective cover **130**. Because strands **80C** are wedged between portions of protective cover **130**, strands **80C** may be resistant to becoming dislodged from substrate **36**. In addition to holding strands **80C** in place so that component **26** remains attached to fabric **12**, grooves **50** may also be used as a physical guide for aligning component **26** relative to fabric **12** during component insertion and attachment operations. This may be beneficial when inserting and attaching component **26** to fabric **12** without line of sight, for example.

(55) Each strand **80C** may align with an associated pad **40** on component **26**. If desired, pads **40** may be formed from elongated strips of conductive material (e.g., metal) that extend from one edge of substrate **36** to an opposing edge of substrate **36**. This provides a large area with which to form a mechanical and electrical connection between substrate **36** and strands **80C**. The elongated shape of pads **40** may allow conductive material **82** to attach a longer portion of strand **80C** to pad **40**. The connection between pad **40** and strand **80C** may, for example, span across the width of substrate **36**, thereby providing a robust connection between substrate **36** and strand **80C**. This is, however, merely illustrative. If desired, pads **40**, conductive material **82**, and the exposed conductive portions of strands **80C** may span across less than all of the width of component **26**.

(56) If desired, component **26** may include internal portions that are located within fabric **12** and external portions that are located on the outside of fabric **12**. To allow mechanical and/or electrical connections between the internal and external portions of component **26**, component **26** may include one or more protrusions that pass through fabric **12** to interconnect the internal and external portions of component **26**. The internal portion of component **26** that is located within fabric **12** may be inserted into fabric **12** during the formation of fabric **12**, as described in connection with FIG. **4**. Following formation of fabric **12**, the external portion of component **26** may be coupled to the internal portion via protrusions on component **26** that pass through fabric **12**.

(57) FIG. **9** is a cross-sectional side view of illustrative fabric **12** having a component with an internal portion located inside of the fabric and protrusions that allow the internal portion to be

coupled to an external portion outside of the fabric. As shown in FIG. 9, fabric 12 may include upper and lower fabric portions 12-1 and 12-2. As discussed in connection with FIG. 4, fabric portions 12-1 and 12-2 may be formed from interlaced strands 80. For example, a first set of strands 80 may be used to form fabric portion 12-1 and a second set of strands 80 may be used to form fabric portion 12-2. Fabric portions 12-1 and 12-2 may be different portions of a single layer of fabric 12, or fabric portion 12-1 may form one or more first layers of fabric 12 and fabric portion 12-2 may form one or more second layers of fabric 12.

(58) Interlacing equipment 120 of FIG. 4 may be used to create one or more regions in fabric 12 such as pocket 66 for receiving electrical components. In the example of FIG. 9, internal portion 26A of component 26 may be inserted into pocket 66 during the formation of fabric 12 (e.g., using component insertion equipment such as insertion tool 54 of FIG. 4). If desired, internal portion 26A may be electrically and/or mechanically connected to one or more strands in fabric 12 such as conductive strand 80C. Internal portion 26A may have grooves of the type shown in FIG. 8 for receiving strands such as strand 80C to provide a more robust electrical and mechanical connection between internal portion 26A and fabric 12. This is, however, merely illustrative. If desired, internal portion 26A may be coupled to an insulating strand 80 (e.g., without being electrically coupled to fabric 12) or may not be connected to any strands 80 (e.g., internal portion 26A may be contained within pocket 66 without being mounted to any strands).

(59) Internal portion 26A of component 26 may include one or more electrical devices 28A. Electrical device 28A may include any of the circuitry described in connection with FIG. 5. In one illustrative arrangement, device 28A of internal portion 26A includes control circuitry for controlling one or more electrical devices in an external portion of component 26 (not shown in FIG. 9). This is merely illustrative, however. If desired, the circuitry in internal portion 26A of component 26 may operate independently of any circuitry in an external portion of component 26.

(60) To provide an electrical and/or mechanical connection between the internal and external portions of component 26, component 26 may include protruding portions such as protruding portions 54 that pass through fabric 12-2. Protruding portions 54 may be leads, pins, posts, flexible wires, and/or any other suitable structures extending from within fabric 12 (e.g., in pocket 66) to the exterior of fabric 12 (e.g., to be exposed on an outer surface of fabric 12). Protrusions 54 may be formed from conductive material to allow electrical signals to pass between the internal and external portions of component 26, or protrusions 54 may be used exclusively for mechanical connection between the internal and external portions of component 26.

(61) Protrusions 54 may, for example, be formed as part of internal portion 26A of component 26. When internal portion 26A is inserted into fabric 12 during the formation of fabric 12, protrusions 54 may initially be located within fabric 12 (e.g., between fabric portions 12-1 and 12-2. When internal portion 26A is received within pocket 66, protrusions 54 may naturally extend through fabric 12 (e.g., through fabric portion 12-2, in the example of FIG. 9), or protrusions 54 may extend through fabric 12 after internal portion 26A is pressed toward fabric 12 (e.g., during formation of fabric 12 or after formation of fabric 12). Protrusions 54 may extend between adjacent strands 80 (e.g., between gaps that already exist between strands 80), may extend through individual strands 80, and/or may extend through openings that are specifically formed in fabric 12 for receiving protrusions 54. There may be one, two, three, four, or more than four protrusions extending from internal portion 26A through fabric 12.

(62) After formation of fabric 12, internal portion 26A may be contained within fabric 12 and protrusions 54 may be exposed on the exterior of fabric 12. This allows an external portion of component 26 to be coupled to the internal portion of component 26 after fabric 12 is formed, as shown in FIG. 10.

(63) As shown in FIG. 10, external portion 26B of component 26 may overlap and be coupled to internal portion 26A by attaching external portion 26B to protrusions 54. External portion 26B of component 26 may include electrical devices 28B. Devices 28B may include any of the circuitry

described in connection with FIG. 5. For example, devices **28B** may include circuitry that is controlled by or otherwise in electrical communication with devices **28A** of internal portion **26A**.

(64) By forming part of component **26** inside of fabric **12** and another part of component **26** outside of fabric **12**, input-output components that operate better outside of fabric (e.g., optical components such as optical sensors, cameras, displays, indicators, speakers, microphones, removable and/or replaceable batteries, haptic output devices, etc.) may be formed on an exterior surface of fabric **12**, while internal circuitry (e.g., control circuitry, logic boards, printed circuits, driver circuitry, etc.) may be hidden or otherwise contained within an interior portion of fabric **12**. This allows signals to be emitted from external portion **26B** of component **26** and/or detected by external portion **26B** of component **26** without interference from fabric **12**. For example, external portion **26B** of component **26** may emit signals in direction **88** towards object **70** and/or may detect signals coming from object **70** in direction **90** without interference from fabric **12**. Signals that are emitted and/or detected by external portion **26B** of component **26** may include optical signals, radio-frequency signals, haptic signals, ultrasonic signals, sound signals, capacitive signals, and/or any other suitable signals.

(65) In one illustrative arrangement, fabric **12** may form a fabric band that is worn around a user's wrist or other body part. External portion **26B** may be an optical component such as a heart-rate sensor (e.g., a photoplethysmography sensor), a blood oxygen sensor, or other optical sensor that rests against a user's skin (or a non-optical heart-rate sensor such as electrocardiographic electrodes that rest against the skin). Internal portion **26A** may include control circuitry for controlling the optical component(s) in external portion **26B** and/or for monitoring output from the optical component(s) in external portion **26B** and taking suitable action (e.g., providing output associated with sensor data gathered by the optical component(s) in external portion **26B**). By mounting control circuitry in internal portion **26A** within fabric **12** and skin-facing sensors in external portion **26B** outside of fabric **12**, sensors may be better positioned for accurate heart rate measurements while a bulk of the circuitry associated with the sensors may be hidden from view within fabric **12**. This is merely illustrative, however. In general, any suitable circuitry may be located within internal and external portions of component **26**.

(66) The arrangement of FIG. **10** in which internal portion **26A** and external portion **26B** have a similar lateral footprint is merely illustrative. If desired, internal portion **26A** may have a smaller or larger lateral footprint than external portion **26B**.

(67) External portion **26B** of component **26** may include mating attachment structures (e.g., recesses, mating pins, solder pads, screw holes, etc.) configured to mate with protrusions **54** to provide an electrical and/or mechanical connection between internal portion **26A** and external portion **26B**. This is merely illustrative. If desired, protrusions **54** may be formed on external portion **26B** of component **26**, and internal portion **26A** of component **26** may include mating attachment features that mate with protrusions **54**. In this type of configuration, protrusions on external portion **26B** of component **26** may be attached to internal portion **26A** of component **26** by inserting the protrusions on external portion **26B** through fabric **12** after fabric **12** is formed to connect to internal portion **26A**.

(68) If desired, an adhesive layer such as adhesive layer **52** may be used to attach external portion **26B** to the exterior surface of fabric **12** (e.g., fabric portion **12-2** in the example of FIG. **10**). Adhesive layer **52** may serve as a watertight seal that prevents moisture and/or other contaminants from entering into the connection region between internal and external portions of component **26**. In arrangements where external portion **26B** is removable (e.g., when external portion **26B** is a removable battery or other removable component), adhesive **52** may be omitted.

(69) In addition to or instead of adhesive between external portion **26B** and fabric **12**, an encapsulating border may be formed around external portion **26B** to seal the connection between external portion **26B** and fabric **12**. This type of arrangement is illustrated in FIG. **11**.

(70) As shown in FIG. **11**, encapsulating border **106** may be located along an outer perimeter of

external portion **26B** and may form a protective ring around external portion **26B**. Encapsulating border **106** may be formed from thermosetting material, thermoplastic material, adhesive, epoxy, and/or any other suitable material for preventing moisture and/or other contaminants from reaching the electrical and/or mechanical connection between external portion **26B** of component **26** and the internal portion **26A** of component **26**.

(71) As shown in the top view of FIG. **12**, encapsulating border **106** may form a rectangular ring around external portion **26B** of component **26**. This is merely illustrative, however. If desired, encapsulating border **106** may have non-rectangular shapes and/or may only partially surround external portion **26B** of component **26**.

(72) FIGS. **13**, **14**, and **15** are side views showing illustrative shapes of protrusions **54**. Protrusions **54** are illustrated as being part of internal portion **26A**, but may be part of external portion **26B**, if desired.

(73) In the example of FIG. **13**, protrusions **54** have end portions **54P** with spherical shapes (e.g., balls). External portion **26B** of component **26** may have mating recesses with shapes that are specifically designed to receive ball-shaped protrusions **54P** of FIG. **13**.

(74) In the example of FIG. **14**, protrusions **54** have end portions **54P** with T shapes. External portion **26B** of component **26** may have mating recesses with shapes that are specifically designed to receive T-shaped protrusions **54P** of FIG. **14**.

(75) In the example of FIG. **15**, protrusions **54** have end portions **54P** with L shapes. External portion **26B** of component **26** may have mating recesses with shapes that are specifically designed to receive L-shaped protrusions **54P** of FIG. **15**. The examples of FIGS. **13**, **14**, and **15** are merely illustrative. Other shapes may be used, if desired.

(76) FIG. **16** shows how external portion **26B** of component **26** may have a surface **56** with recesses such as curved recesses **58** for receiving protrusions **54** of internal portion **26A** of component **26**. Providing recesses **58** with a curved shape allows for a robust mechanical connection in which end portions **54P** of protrusions **54** are first inserted into wider end regions **58P** of recesses **58** and subsequently rotated in directions **60**, thereby locking external portion **26B** into place on internal portion **26A**. Wider end regions **58P** may be shaped to receive the wider end portions **54P** of protrusions **54**, while the rest of recess **58** may be sufficiently narrow such that wider end portions **54P** are unable to pass through. When it is desired to remove external portion **26B** from internal portion **26A**, external portion **26B** may be rotated in the opposite direction so that each end portion **54P** can slide within recess **58** to exit recess **58** at end regions **58P**.

(77) FIG. **17** is a cross-sectional side view showing an illustrative example in which internal portion **26A** of component **26** is coupled to upper and lower external portions **26B** and **26C**.

(78) As shown in FIG. **17**, a first external portion such as external portion **26B** of component **26** may be coupled to internal portion **26A** by attaching external portion **26B** to a first set of protrusions **54** on a first side of internal portion **26A** (e.g., a side facing fabric portion **12-2**). A second external portion such as external portion **26C** of component **26** may be coupled to internal portion **26A** by attaching external portion **26C** to another set of protrusions **54** on a second opposing side of internal portion **26A** (e.g., a side facing fabric portion **12-1**). If desired, adhesive **52** may be used to attach external portion **26C** to an exterior surface of fabric portion **12-1**.

(79) External portion **26B** of component **26** may include one or more electrical devices **28B**, and external portion **26C** of component **26** may include one or more electrical devices **28C**. Devices **28B** and **28C** may include any of the circuitry described in connection with FIG. **5**. For example, devices **28B** and **28C** may include circuitry that is controlled by or otherwise in electrical communication with devices **28A** of internal portion **26A**.

(80) The arrangement of FIG. **17** allows signals to be emitted from external portions **26B** and **26C** of component **26** and/or detected by external portions **26B** and **26C** of component **26** without interference from fabric **12**. For example, external portion **26B** of component **26** may emit signals in direction **96** towards object **72** and/or may detect signals coming from object **72** in direction **98**

without interference from fabric **12**. External portion **26C** of component **26** may emit signals in direction **92** towards object **74** and/or may detect signals coming from object **74** in direction **94** without interference from fabric **12**.

(81) Signals that are emitted and/or detected by external portion **26B** and/or external portion **28C** of component **26** may include optical signals, radio-frequency signals, haptic signals, ultrasonic signals, sound signals, capacitive signals, and/or any other suitable signals.

(82) In one illustrative arrangement, fabric **12** may form a fabric band that is worn around a user's wrist or other body part. External portion **26B** may be a skin-facing input-output device (e.g., an optical component such as a heart-rate sensor, a blood oxygen sensor, or other optical sensor that rests against a user's skin, a non-optical heart-rate sensor such as electrocardiographic electrodes, a haptic output device that provides haptic output to a user, etc.), while external portion **26C** may be a viewer-facing input-output device such as a display, an indicator, a camera, a button, a rotating knob, a touch screen or other touch sensor, a fingerprint sensor, and/or any other suitable input-output component that the user may want to look at during use.

(83) Internal portion **26A** may include control circuitry for controlling the input-output component(s) in external portions **26B** and **26C** and/or for monitoring output from the input-output component(s) in external portions **26B** and **26C** and taking suitable action. By mounting control circuitry in internal portion **26A** within fabric **12** and skin-facing or viewer-facing input-output components in external portions **26B** and **26C** outside of fabric **12**, input-output components may be better positioned for the desired input-output capabilities without interference from fabric **12**, while a bulk of the circuitry associated with the input-output components may be hidden from view within fabric **12**. This is merely illustrative, however. In general, any suitable circuitry may be located within internal and external portions of component **26**.

(84) In some arrangements, fabric **12** with electrical components having internal and external portions may be used to form a strap for a wearable device such as a watch. This type of arrangement is illustrated in FIG. **18**. Device **10** may be worn on a body part of a user (e.g., the user's wrist, arm, head, leg, or other portion of the user's body). As an example, device **10** may include a wearable band, such as band **76** of FIG. **18**. Band **76**, which may sometimes be referred to as a wristband, wrist strap, or wristwatch band, may be formed from polymer, metal, fabric, leather or other natural materials, and/or other material, may have links, may stretch, may be attached to housing **68** in a fixed arrangement, may be detachably coupled to housing **68**, may have a single segment or multiple segments joined by a clasp, and/or may have other features that facilitate the wearing of device **10** on a user's wrist. Housing **68** may be formed from polymer, metal, glass, crystalline material such as sapphire, ceramic, fabric, fibers, fiber composite material, natural materials such as wood and cotton, other materials, and/or combinations of such materials.

(85) Device **10** may include one or more displays such as display **62**. The displays may, for example, include an organic light-emitting diode display, a liquid crystal display, a display having an array of pixels formed from respective light-emitting diodes (e.g., a pixel array having pixels with light-emitting diodes formed from respective crystalline light-emitting diode dies such as micro-light-emitting diode dies), and/or other displays. The displays may include rigid display structures and/or may be flexible displays. For example, a light-emitting diode display may have a polymer substrate that is sufficiently flexible to be bent. Display **62** may have a rectangular pixel array or a pixel array of another shape for displaying images for a user and may therefore sometimes be referred to as a pixel array. Display **62** may also sometimes be referred to as a display panel, display layer, or pixel layer. Each pixel array in device **10** may be mounted under a transparent housing structure (sometimes referred to as a transparent display cover layer, protective cover layer structures, etc.).

(86) FIG. **19** is a cross-sectional side view of device **10** of FIG. **18**. In the example of FIG. **19**, device **10** is worn around a user's body part such as wrist **78**. Strap **76** may extend partially or entirely around wrist **78** and may include one or more clasps, buckles, or other attachment

mechanisms for removable coupling device **10** to wrist **78**. Strap **76** may be formed from fabric **12** and may include one or more electrical components such as component **26**. Component **26** may include an internal portion such as internal portion **26A** located within fabric **12** and one or more external portions such as external portion **26B** located on an exterior surface of fabric **12** (e.g., as described in connection with FIGS. **9-17**). Internal portion **26A** may be inserted into fabric **12** during the formation of fabric **12** and may have protrusions **54** that protrude through fabric **12**. External portion **26B** may be coupled to internal portion **26A** after fabric **12** is formed. External portion **26B** may have mating attachment structures such as mating recesses and/or mating protrusions that mate with protrusions **54** to mechanically and/or electrically couple external portion **26B** to internal portion **26A**. Adhesive **52** (and/or an encapsulating border such as encapsulating border **106** of FIGS. **11** and **12**) may be used to create a seal between external portion **26B** and fabric **12**.

(87) In the example of FIG. **19**, external portion **26B** may form a back crystal module of device **10** with one or more skin-facing optical sensors. For example, external portion **26B** may include one or more light-emitting devices **182** (e.g., one or more light-emitting diodes or lasers operating at visible and/or infrared wavelengths) and/or one or more light detectors **84** (e.g., a photodiode and/or other suitable light sensor). Light-emitting device **182** may emit light towards wrist **78** and light detector **84** may detect light that is reflected back from wrist **78**. Internal portion **26A** of component **26** may include control circuitry such as control circuitry **86**. Control circuitry **86** may send control signals to light-emitting diode **182** and/or light detector **84** and/or may receive sensor data from light-detector **84** via protrusions **54**.

(88) If desired, housing **68** and display **62** may be formed on an opposing surface of fabric **12**, so that viewer **102** can view display **62** in direction **104**. Housing **68** and display **62** may, if desired, be electrically connected to internal portion **26A** (e.g., via conductive strands **80C** in fabric **12**, via conductive protrusions **54** on an upper side of internal portion **26A** as in the example of FIG. **17**, and/or via other conductive structures in device **10**). This is, however, merely illustrative. If desired, housing **68** and display **62** may not be electrically connected to component **26** (e.g., data may be wirelessly transferred between circuitry in housing **68** and circuitry in component **26**).

(89) As described above, one aspect of the present technology is the gathering and use of information such as information from input-output devices. The present disclosure contemplates that in some instances, data may be gathered that includes personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can include demographic data, location-based data, telephone numbers, email addresses, twitter ID's, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, username, password, biometric information, or any other identifying or personal information.

(90) The present disclosure recognizes that the use of such personal information, in the present technology, can be used to the benefit of users. For example, the personal information data can be used to deliver targeted content that is of greater interest to the user. Accordingly, use of such personal information data enables users to have control of the delivered content. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure. For instance, health and fitness data may be used to provide insights into a user's general wellness, or may be used as positive feedback to individuals using technology to pursue wellness goals.

(91) The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for

legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the United States, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA), whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

(92) Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, the present technology can be configured to allow users to select to “opt in” or “opt out” of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select not to provide certain types of user data. In yet another example, users can select to limit the length of time user-specific data is maintained. In addition to providing “opt in” and “opt out” options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an application (“app”) that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

(93) Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user's privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data at a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.

(94) Therefore, although the present disclosure broadly covers use of information that may include personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data.

(95) The foregoing is merely illustrative and various modifications can be made by those skilled in the art without departing from the scope and spirit of the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. A fabric item, comprising: first and second fabric portions that overlap one another and that form a closed pocket; an electrical component, comprising: an internal portion located in the closed pocket between the first and second fabric portions; an external portion located on an exterior surface of the second fabric portion, wherein the external portion overlaps the internal portion and

wherein the second fabric portion is interposed between the internal portion and the external portion; and protrusions that extend through the second fabric portion and that couple the internal portion to the external portion; and adhesive interposed between the external portion and the second fabric portion.

2. The fabric item defined in claim 1 wherein the protrusions comprise a conductive material that conveys electrical signals between the internal portion and the external portion.

3. The fabric item defined in claim 1 wherein the external portion comprises an input-output device and the internal portion comprises control circuitry that is electrically coupled to the input-output device via the protrusions.

4. The fabric item defined in claim 3 wherein the input-output device comprises an optical sensor having a light-emitting diode and a light detector.

5. The fabric item defined in claim 4 wherein the optical sensor is selected from the group consisting of: a heart-rate sensor and a blood oxygen sensor.

6. The fabric item defined in claim 1 further comprising a conductive strand that passes between the first and second fabric portions, wherein the internal portion of the electrical component is mounted to the conductive strand.

7. The fabric item defined in claim 1 wherein the protrusions extend from the internal portion and wherein the external portion comprises recesses that receive the protrusions.

8. The fabric item defined in claim 7 wherein the recesses have a curved shape and wherein the protrusions are configured to slide within the recesses as the external portion is rotated relative to the internal portion.

9. The fabric item defined in claim 1 wherein the first and second fabric portions form a wrist strap.

10. The fabric item defined in claim 1 further comprising an encapsulating border located on the exterior surface of the second fabric portion that surrounds a periphery of the external portion.

11. A strap, comprising: fabric formed from interlaced strands of material, wherein the fabric has first and second fabric portions that overlap one another and that form a closed pocket; and an electrical component having a first portion located inside the closed pocket of the fabric, a second portion located outside of the fabric overlapping the first portion, and pins that extend through the fabric to couple the first portion to the second portion, wherein the second portion comprises a skin-facing optical sensor and the first portion comprises control circuitry that electrically communicates with the skin-facing optical sensor via the pins.

12. The strap defined in claim 11 wherein the skin-facing optical component comprises a light-emitting diode and a light detector.

13. The strap defined in claim 12 wherein the skin-facing optical component is selected from the group consisting of: a heart-rate sensor and a blood oxygen sensor.

14. The strap defined in claim 11 further comprising an adhesive interposed between the second portion and the fabric.

15. The strap defined in claim 11 wherein the strands of material comprise a conductive strand and wherein the first portion is mounted to the conductive strand inside of the fabric.

16. An electronic device, comprising: fabric formed from interlaced strands, wherein the fabric has first and second fabric portions that overlap one another and that form a closed pocket; and an electrical component comprising: an internal portion located in the closed pocket; an external portion uncovered by the fabric and exposed on an exterior surface of the fabric overlapping the internal portion; and pins that pass through the fabric to electrically couple the internal portion to the external portion.

17. The electronic device defined in claim 16 wherein the external portion has recesses that receive the pins.

18. The electronic device defined in claim 17 wherein the external portion is removably coupled to the internal portion by rotating the external portion relative to the internal portion.

19. The electronic device defined in claim 18 wherein the external portion comprises a removable battery.
